Validation Windows: Epistemic Uncertainty Produced by the Solver in a Literature-Derived Bayesian Network

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Abstract

When a Bayesian network is built from published scientific studies, each contributing a point estimate and a 95% confidence interval, the aggregation problem is fundamentally epistemic, not just statistical. The literature on a node's different parents can disagree over the joint cells they share; the law of total probability ties those cells together; and the resulting conditional probability tables (CPTs) must also be consistent with observed population frequencies. We describe a system that solves the aggregation as a quadratic program whose feasible region is a convex polytope of CPTs consistent with all sources of evidence simultaneously. How far each study's literature band must be scaled for the network to stay jointly feasible (which we call the validation window) is a first-class representation of epistemic uncertainty for that edge: produced by the construction itself, not computed post-hoc as a calibration layer. It is tight where literature and population data agree, wide where they disagree, and exceeds the study's own literature CI when the solver must stretch the study to achieve global consistency. The system sits between two transformer endpoints: an upstream LLM that extracts literature into structured spreadsheet rows (making the pipeline domain-portable), and a downstream multi-agent coevolutionary simulation it seeds with behaviors. On a 573-node longevity-risk network built from 193 published studies, the median validation window is 0.001; a controlled evidence-addition demonstration on an earlier build shows the window shrinking by 87% (from 0.125 to 0.016) as consistent literature-established edges are added. A residual set of low-base-rate edges where the solver cannot commit to a direction, the $\approx 2\times10^{-4}$ pancreatic-cancer chain, demonstrates the intended failure mode: abstention, visible as inversion of the CPT direction precisely where literature alone cannot discipline the joint. Beyond the window, the construction is also usable as a predictor: on the current build, with each target's definitional-surrogate biomarkers excluded, it discriminates held-out clinical events at a mean per-respondent ROC AUC of 0.77 across 12 of its 15 constructed disease targets (up to 0.95 on coronary artery disease), at or above the level of cohort-fitted clinical risk calculators on the better-instrumented targets.

1. Introduction

A Bayesian network built from the published scientific literature faces three simultaneous epistemic challenges. Individual uncertainty: each primary or meta-analytic study reports a point estimate and an interval, not a single exact value. Cross-edge disagreement: a node's parents carry separate literatures, and those marginals can imply conflicting values for the joint CPT cells they share once the law of total probability couples them. Population coherence: the assembled CPTs must remain consistent with population-level frequencies from a reference survey dataset. Collapsing all of this to a single CPT per node, the default in published expert systems (Heckerman, 1998), discards most of the epistemic information before it reaches the user. Returning the set of CPTs consistent with all sources, by contrast, preserves the shape of what is not known.

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Transformers sit on both ends of the pipeline we describe. Upstream, a large language model (Brown et al., 2020) reads published meta-analyses and primary studies, extracting relative risks and other effect sizes, confidence intervals, populations, and citations into the structured spreadsheet rows our solver consumes. The LLM also supplies the causal structure a population survey cannot: edge directions and mediating-node placement, from the literature's mechanistic consensus (App. D). This LLM-mediated extraction is what makes the method generalizable: any scientific domain with peer-reviewed effect-size literature, longevity and health in our demonstration, but also macroeconomic policy, education interventions, or criminal-justice evaluation, can be converted to a network without domain-specific hand-curation. Downstream, once the network carries treatment (interventional) edges it is itself a policy net that seeds the agents of a multi-agent coevolutionary simulation with behaviors.

This paper presents the network-construction step. The system takes as input a spreadsheet of literature-sourced edges (relative risks and other effect sizes plus 95% confidence intervals) together with a reference dataset of population covariate frequencies (we use NHANES (Centers for Disease Control and Prevention, National Center for Health Statistics, 2020) throughout), and produces a Bayesian network whose CPTs are the solution of a quadratic program. The QP's objective is heuristic, a least-squares fit to the literature point estimates plus a monotonicity penalty that discourages direction inversions, and its constraints are probability-law: the CPT is a valid probability table, each literature RR lies within a windowed band, and the law of total probability holds across groups of cells. We call the last item the subset constraints. The feasible region of the constrained QP is a convex polytope of CPTs consistent with all sources of evidence simultaneously.

Our contribution is conceptual and algorithmic: a working quadratic-program solver that constructs the polytope from literature in a single pass, together with the proof (Theorem 3) and argument that the scaling each study's band must take for the constrained solve to stay feasible, the validation window, is the quantity the system already computes that directly answers the core epistemic-uncertainty question: for this edge, how much does the model not know? Tight windows mean "literature and population data mutually confirm this effect under the structure I've built." Wide windows mean "I had to bend the literature to make the whole thing fit." Windows that exceed the raw literature CI mean "my structure may be wrong; attend here." The system's operating loop iteratively searches for structural changes or additional literature that narrows the widest windows; the median window serves as the monotone-decreasing measure of residual epistemic uncertainty. When the network is handed off to downstream transformer agents, these windows travel with each edge: the transformer receives a prior that makes its own uncertainty commensurate with the literature's. As supplementary evidence that these literature-constrained CPTs carry real predictive structure, the network discriminates held-out clinical events at a mean per-respondent ROC AUC of 0.77 (up to 0.95), with diagnostic biomarkers excluded (App. J).

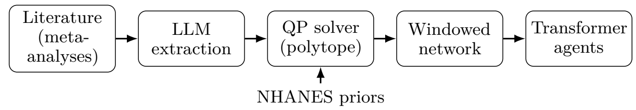

Figure 1. The end-to-end pipeline. An LLM extracts published effect sizes into structured rows; the QP solver returns the feasible polytope of CPTs; each edge carries its validation window into the downstream transformer agents. NHANES supplies per-node priors to the solver.

The system is not an imprecise-probability engine (§3 relates the feasible polytope to credal sets (Walley, 1991; Cozman, 2000)). It is not a calibration (Guo et al., 2017) or conformal-prediction (Vovk et al., 2005) overlay on a pre-trained classifier. It is a construction process whose intermediate artifact, the windowed polytope, is the epistemic representation. The construction itself reports the residual uncertainty on each edge.

2. The System End-to-End

Figure 1 summarises the pipeline end to end; the three construction stages (Inputs, Compilation, CPT solve) are described in turn.

Inputs. A single spreadsheet. A study row carries an output node (e.g. lung_cancer), an input node and its value (smoking_yes), a statistic type (RR, OR, HR, or SMD), a point estimate, a ± for the 95% CI, and a citation; the network draws on 193 published studies, predominantly meta-analyses. A definition row declares a node's type: a directly-observed NHANES variable, a dependency node whose CPT the QP solves from its parents' study RRs, or a logical aggregator (any_of, all_of) combining parents. Still other rows declare a link between nodes: a subset relation (is_a, subsumes) or identity (equivalent_to) for definitional containment, or a distal link routing a computed aggregator-level effect onto a disease. The build also takes the reference dataset, the full continuous NHANES (1999–2020), 116K respondents.

Compilation. The spreadsheet compiles to a DAG with one node per output and one edge per (input, output) pair. Aggregator definitions resolve to deterministic truth tables. Distal links, organizing leaves into sub-aggregators that feed a disease, are handled by a Monte-Carlo procedure that propagates study-level RRs through the aggregator's NHANES-weighted co-occurrence structure to produce a single aggregator-to-disease RR with its own 95% CI.

CPT solve. For each node Y with parents $X_1 \dots X_K$, let $\mathbf{x}$ be the flattened CPT. Each literature study i bearing on Y gives four linear constraints $r_k^{(i)}\mathbf{x} = s_k^{(i)}$ (the standard marginal-RR expansion): each $r_k^{(i)}$ expresses study i's marginal RR as a linear functional of the flattened CPT via the law of total probability, fixed by the study's (parent, value, target) and the reference-survey marginal. Stacking these rows gives the matrix R (the collection of the $r_k^{(i)}$) and target vector $\mathbf{s}$. The QP objective (both terms heuristic) is

$$\mathcal{L}(\mathbf{x}) = \|R\mathbf{x} - \mathbf{s}\|_2^2 \; + \; \lambda \sum_p \max(0, \mathrm{inv}_p)^2, \qquad (1)$$

where the first term fits the literature point estimates and the second penalises monotonicity inversions, with weight $\lambda > 0$ (p indexes signed-direction (parent, value) pairs and inv_p is the violation amount, zero when the direction holds). We do not claim this objective follows from a Bayesian or decision-theoretic principle; it is a simple heuristic that merely selects one interior point of the polytope. The probability laws live in the constraints, not the objective, so the reported window w depends only on the constraint polytope (§3, App. A) and is invariant to λ. The constraints that encode probability laws are: (i) $\mathbf{x}$ is a valid probability table (row sums, non-negativity); (ii) a windowed literature band per study, $s - w\,\mathrm{CI} \le R\mathbf{x} \le s + w\,\mathrm{CI}$, with CI the study's own literature 95% half-width (narrower CIs yield tighter bands) and w a per-edge scaling determined by a binary search on feasibility; (iii) subset constraints, a set of linear equalities/inequalities requiring that, for every group of CPT cells linked by the law of total probability, the corresponding relation holds within the windowed tolerance. Subset constraints range over groups of cells induced by the CPT's full structure, including marginalisation over any subset of parents. Grouping the terms this way isolates the heuristic part of the solve (literature-fit and monotonicity) from the probability-law part (the constraints). Least squares and monotonicity are retained as soft heuristics because marginal effect sizes and their directions are known properties of the literature, not probability laws. NHANES supplies per-node priors through the definition rows; it does not supply conditional training signal to the solver, which would create a circularity in the evaluation metrics below.

The validation window w scales the confidence band. The polytope's boundary is determined by the probability-law constraints, not by the objective; w is therefore independent of λ. The window w is not a free parameter; it is the smallest scaling at which the constrained QP stays feasible, found by binary search. $w{=}1$ places the band at the literature CI: consistent cross-evidence drives w below 1, irreconcilable evidence above 1 (the band stretched past the CI). $w{=}0$ means the cell is uniquely determined; an edge that settles at $w{=}1$ is held by its study's CI alone.

Under the normal approximation, each input 95% CI is a likelihood-ratio confidence region at level $\alpha{=}0.05$, so the per-edge band $s \pm w\,\mathrm{CI}_e$ is a confidence interval for the edge's CPT cell, with a per-edge coverage cov_e reported alongside. Coverage scales with a Bonferroni multiplier c via $c\,z_{1-\alpha/2} = z_{1-\alpha/(2n)}$; for the demonstration's $n{=}166$ studies, joint coverage 0.95 corresponds to $c{\approx}1.85$. Runs in this paper use $c{=}1$; intervals are exact marginally and conservative jointly, and any target joint coverage can be regenerated via the Bonferroni relation. When studies on different edges are mutually consistent, w collapses toward 0 and the band tightens well inside the literature CI; when they conflict, w must widen, and an edge forced past $w{=}1$ (its band stretched beyond the full literature CI) is the system's explicit complaint that its literature will not reconcile with the network.

3. Windows as Epistemic Uncertainty

The feasible polytope of CPTs satisfying all constraints at the minimum-feasible w is a convex region (Fig. 2). Along each study axis its extent is $[L_i, U_i]$, and the width $U_i - L_i \in [0, 1]$ is the edge's feasible range r_i, the polytope projection the window mechanism also yields. By construction this is a projected credal set in the sense of Walley (1991) and Cozman (2000), but arrived at constructively from published literature rather than from elicited lower and upper probabilities. Four properties make w a useful epistemic report:

(1) Monotone in evidence. Adding a consistent literature edge can only tighten the polytope (Theorem 3, part 1), and a binding one strictly tightens it (part 2); an inconsistent edge is detected because w must widen (or the QP becomes infeasible). In our experiments

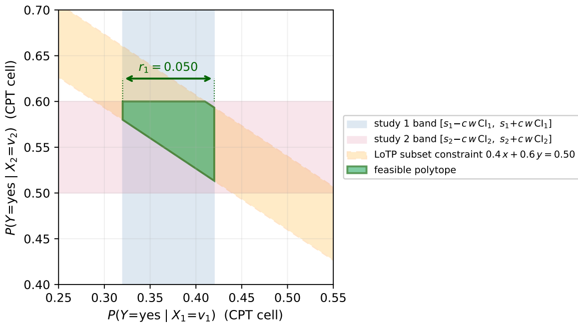

Figure 2. Feasible polytope of CPTs from two studies and a law-of-total-probability (LoTP) subset constraint. Axes are CPT-cell probability units, e.g. $P(Y\text{=yes} \mid X_1\text{=}v_1)$ on the horizontal axis (each study's literature RR multiplied by the base rate gives a band on its corresponding axis). Blue and pink: the two literature CI bands. Orange dashed: the LoTP subset-constraint band. Green polygon: the feasible polytope (solution set); its projection onto the study-1 axis is the edge's feasible range r_1. When the projection equals the literature CI the system has extracted no extra information; when narrower, cross-CPT coherence has tightened the prior; when wider, the solver has stretched the study to keep the QP feasible, signalling cross-study tension.

the median window $\tilde{w}$ dropped by 87% (from 0.125 to 0.016) as eight structural edges from a medical-knowledge audit were added, meaning that by the final build, half of all solved cells have $w \leq 0.016$ (their literature + subset-constraint content is almost fully binding), and the remaining half spread up to 1.0.

(2) Localizes cross-edge tension. The system sorts studies by w. Studies with high w are edges where the per-study literature CI disagrees with the joint of all other constraints in the network: either literature on this edge contradicts literature on an adjacent edge, or literature contradicts population data, or the DAG is missing a structural edge that would resolve the tension. In the demonstration network the top-20 window list after each iteration provided the to-do list for the next iteration. The three edges that remain stuck at w=1 after all available evidence has been added are the diabetes/healthy-diet pairs; their literature pulls in a direction that no current structural change has reconciled with the adjacent constraints.

(3) Exceeds the literature CI when forced. A window wider than the study's own literature CI is a report by the system that something upstream is wrong: the chosen structure, another study, or a NHANES leaf mapping.

(4) Determines untested parent combinations. For K binary parents the CPT has 2^K rows (binary parents assumed for exposition, as most of ours are); literature typically measures only the K marginal effects. Given the marginals and the reference survey's parent co-occurrence structure, the subset constraints mathematically determine the remaining $2^K - K$ rows to within the window, combinations no study has jointly examined, provided the co-occurrence exceeds a threshold we make precise in App. C. We measure co-occurrence with a signed deviation, $\rho(X_i, X_j) = \max_{v_i}\left[P(X_j = \hat{x}_j \mid X_i = v_i) - P(X_j = \hat{x}_j)\right]$, with $\hat{x}_j$ the parent's reference state and the sign of the maximising deviation retained. Because the derivation is the law of total probability, these predictions need no further experiment. The required $|\rho|$ is modest; where it is not yet sufficient, the objective returns the best estimate given the known marginals and the DAG, with the window reporting the residual uncertainty.

A complementary abstention signature (§5) is direction symmetry within a small feasible polytope: w can be modest yet the solver cannot pick a committed direction. That failure does not surface in w; it surfaces in the query-RR direction check.

4. Empirical Demonstration

Setup. The evidence-addition demonstration of this section uses an earlier 427-node, 166-study build (the canonical 573-node, 193-study build's held-out AUC, objective-RR fidelity, and calibration are in App. J), anchored to the full continuous NHANES (1999–2020), 116K respondents. On the demonstration build we evaluate three things:

- Direction accuracy: for each cited edge, clamp its input at yes, query its target, compare the sign of the resulting query RR to the literature RR's sign; report the fraction correct out of 369 testable cited edges (a local clamp; App. J gives the propagated full-chain 85.4%).
- Median window $\tilde{w}$: median of w over the per-cell CPT rows of the 287 solved nodes.
- Joint fidelity within 5%: exhaustive over every CPT cell for which a NHANES ground truth exists, i.e., every parent of the node and the node itself carry NHANES codes, and $\geq$ 30 NHANES respondents match the parent combination. Latent aggregators and distal nodes are excluded only because they are not directly observed (no ground truth), not by selection. Under this rule the demonstration network yields 178 cells covering 15 nodes, rising to 194 cells / 16 nodes after the final evidence-addition row (which introduces a new all-NHANES-observable node). We report the fraction with

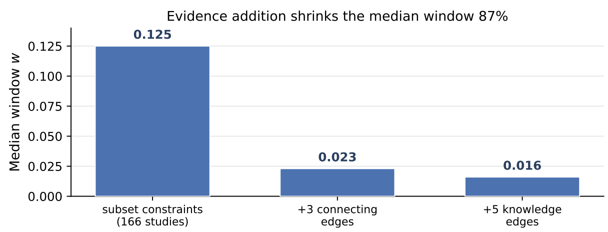

Figure 3. Median validation window across the evidence-addition iterations: $\tilde{w}$ shrinks 87% (from 0.125 to 0.016) as consistent literature edges are added. Data from Table 1.

$$|\text{CPT} - \text{NHANES-empirical}| < 0.05.$$

Cycles of evidence addition. Table 1 reports the median window $\tilde{w}$ and direction accuracy as structural edges from a medical-knowledge audit are added, and Fig. 3 plots the trajectory. Because the current 573-node build already sits at $\tilde{w} = 0.001$, the window's response to evidence is demonstrated on the earlier 427-node, 166-study build, where the eight added edges collapse $\tilde{w}$ from 0.125 to 0.016, an 87% reduction in epistemic slack. Under the full query-RR test the 573-node build reproduces the study direction on 85.4% of edges (App. J, Table 3), below Table 1's near-98%, which the chain washout (§5) lowers under the propagated query. The structural-audit edges (BMI→diabetes, diabetes→CAD, and six further meta-analysis-backed edges) tighten the polytope by supplying cross-edge coherence that resolves prior network tensions.

The widest-w studies as research agenda. In the final build the twenty widest-w studies cluster into three groups: (a) transportability-limited studies whose exposure definitions do not map cleanly onto NHANES covariates (occupational nickel exposure vs. NHANES urinary nickel); (b) structurally underspecified chains where the disease has few parents and the solver has latitude; (c) correlated co-parents entangled in NHANES but represented as independent in the network. Each group suggests a distinct next-action: find a population-based study, add a directly-observed biomarker, or introduce a dependency constraint between the co-parents. The widest-w list is the system's internally-generated research agenda, deriving next actions from explicit ignorance localization, rather than from a flat confidence score.

5. Abstention at Low Base Rate

Pancreatic cancer has a NHANES prevalence of $\approx 2\times10^{-4}$. Seven studies linking occupational chemical exposures to pancreatic cancer, with literature RRs 1.3–1.7, do not produce the expected direction-correct CPT cells in our solved network. The distal-computed RR (aggregator → disease) is 2.60; but the QP's solution assigns $P(\text{panc}|\text{chem=yes}) < P(\text{panc}|\text{chem=no})$, the opposite direction.

This is not a bug. At base rate 2×10^{-4} the absolute-error landscape of the LS term is vanishing (10^{-7} units), and the monotonicity penalty at the default margin tolerates inversions of magnitude 10^{-2} without penalty. With both soft objective terms effectively flat, the solver's choice is driven by the hard subset constraints alone, and at such low marginal, the law-of-total-probability relations over the CPT cells admit a feasible region nearly symmetric along the parent-direction axis. Any point in that region satisfies the constraints equally well; the solver's converged point falls on the direction-inverted side. This is the correct report that literature plus the law of total probability alone do not uniquely determine the conditional direction at this base rate. In the system's own language it is reporting that it abstains.

The per-study window w does not catch this case: in cycle-12 data the seven pancreatic studies have windows in the mid-range (not in the top-20 widest), because the solver did find a feasible point within each literature CI, it just happened to pick one on the direction-inverted side of a symmetric feasible region. w reports the radius of the feasible region; it does not report the symmetry of that region under reflection through the direction-neutral point. The query-RR direction check (§4) catches it: the seven pancreatic edges all return query RR = 0.055 against literature RR 1.3–1.7, a maximum-signal disagreement. A deployment layer should therefore flag any query whose RR reverses the literature sign as a potential abstention, independent of w, and route the user to the widest-w and direction-inverted studies together with the statement "at this base rate, available literature does not uniquely determine the conditional direction."

We attempted two solver-side remediations, scaling the monotonicity-penalty weight by 10× and tightening the inversion tolerance from 0.01 to 0.0001, and neither dislodged the inversion, confirming that the issue is geometric (the feasible polytope at low base rate is symmetric along the direction axis) and not a matter of tuning. The right treatment is post-solver: at query time, when a queried cell lies in a region of the polytope whose symmetric reflection is also feasible, the response should be the window rather than the point. This is a lifelong-learning operation: when new literature or population data narrow the polytope, the cell resolves; until then, the system knows it does not

Table 1. Evidence-addition iterations. Subset constraints are on throughout. Adding literature-established edges (rows 2–3) reduces $\tilde{w}$ by 87% while holding direction accuracy within 0.5 percentage points; joint fidelity rises from 77.0% to 78.9%. Direction uses a fixed 369-edge set across rows; only the joint denominator grows ($178 \to 194$). The "Median CI coverage" column is $\mathrm{erf}(w\, z_{0.975}/\sqrt{2})$, the coverage of the solved band $s \pm w\,\mathrm{CI}$; it shrinks as evidence is added because $w \ll 1$ makes that band far narrower than the study's 95% CI: the cell is pinned, so small is tight, not deficient. The 95% (marginal, c=1) / joint-95% (c≈1.85) coverage guarantee is a separate property of the input bands $s \pm c\,\mathrm{CI}$ (App. A). A coverage level, not an NHST p-value.

Build	Direction accuracy	Median $\tilde{w}$	Median CI coverage	Joint within 5%
Subset constraints on, 166 studies	362/369 (**98.1**%)	0.125	0.194	137/178 (77.0%)
+ 3 connecting-study edges (e.g. bmi→diabetes, diabetes→CAD)	360/369 (97.6%)	0.023	0.036	137/178 (77.0%)
+ 5 medical-knowledge edges (bmi→hypertension, etc.)	360/369 (97.6%)	**0.016**	**0.025**	153/194 (78.9%)

know.

6. Discussion

The constructed network is also usable as a predictor: it discriminates held-out clinical events at a mean ROC AUC of 0.77 across 12 targets (up to 0.95, biomarkers excluded; App. J). Because the law of total probability couples the studies across edges, the discrimination need not reduce to the studies: the solver fills joint-conditional cells no single study measured, so the coupling itself may add predictive signal. Externally, where the raw NHANES marginal misleads (insomnia→hypertension, 2.83 versus ≈1.2 in the literature, the net, and held-out UK Biobank), the net follows the literature, evidence it adds beyond the population correlations (App. K).

Each study row is an update operator: consistent evidence shrinks the polytope and lowers w, while disagreement forces w wider and surfaces in the widest-w list, so re-solving on every update keeps w a faithful measure of residual disagreement. This affords automatic curation (accept a row if the median window decreases) and hypothesis generation (rank structural changes by polytope narrowing), and the crowd-sourced construction loop of App. N. The spreadsheet rows are LLM-extracted (Brown et al., 2020) from a released manual, making the method portable and its errors auditable as wide windows (App. G).

7. Related Work

Imprecise probability (Walley, 1991) and credal networks (Cozman, 2000; Piatti et al., 2010) formalise uncertainty as a convex set of distributions, with LP-based inference algorithms developed by Antonucci et al. (2015) and surveyed by Mauá and Cozman (2020); recent work carries credal sets into deep uncertainty quantification (Caprio et al., 2024). Our feasible polytope is isomorphic to a separately specified local credal set, but constructed from published literature rather than from elicited lower–upper bounds; the convex set arises as a by-product of a literature-constrained construction rather than being learned or elicited. Dirichlet distributions over CPT rows (Heckerman, 1998) do not couple across edges; our subset constraints do. Calibration (Guo et al., 2017; Kuleshov et al., 2018) and conformal prediction (Vovk et al., 2005; Romano et al., 2019) adjust outputs post hoc; we report the solver's own feasibility interval. Classifier abstention (Hendrycks and Gimpel, 2017; Liu et al., 2020) declines to predict; we abstain where the polytope is symmetric along the direction axis. Retrieval-augmented generation (Lewis et al., 2020) uses literature at inference; we use literature as hard constraints at construction.

The causal-direction layer parallels expert-elicited DAG construction in epidemiological causal inference (Hernán and Robins, 2020), the LLM serving as the domain expert; its dependence- and mediation-handling (App. D) are construction-vocabulary analogues of operations studied by Pearl (1988) and VanderWeele (2015), yielding literature-coherent observational conditionals informed by causal direction, not a claim of Pearl-sense interventional identification.

8. Conclusion

The validation window w is produced by the model-construction procedure itself. On the demonstration network the polytope shrank 87% as evidence was added, now provably monotone (Theorem 3); the widest windows named research directions. Where the solver could not commit (the pancreatic-cancer chain at $2{\times}10^{-4}$) it abstained: a model that knows what it does not know. Code and data at https://github.com/agentbasedlearningsystems/validation-windows.

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Appendix, Supplementary Material

A. Per-edge and joint coverage

We make the coverage claim of §2 precise and state its assumptions, since the window w is meant as a confidence statement. Two assumptions. (A1) Under a normal approximation, each input study's reported 95% CI is a Wald interval for the effect-size parameter on that edge, with endpoints $s \pm z_{1-\alpha/2}\,\sigma$ at $\alpha = 0.05$. (A2) The solver's per-edge input band is that interval scaled by the Bonferroni multiplier c, namely $s \pm c\,z_{1-\alpha/2}\,\sigma$.

Per-edge coverage. Under (A1)–(A2), the input band of half-width $c\,z_{1-\alpha/2}\,\sigma_e$ on edge e has marginal coverage $\mathrm{cov}_e = 2\Phi\big(c\,z_{1-\alpha/2}\big) - 1 = \mathrm{erf}\big(c\,z_{1-\alpha/2}/\sqrt{2}\big)$. At $c = 1$ this is exactly the nominal 0.95.

Joint coverage. Treating n edges as simultaneous and choosing c so that $c\,z_{1-\alpha/2} = z_{1-\alpha/(2n)}$ gives family-wise coverage at least $1 - \alpha$ by the Bonferroni inequality, for any dependence among edges. For the demonstration network ($n = 166$) this is $c \approx 1.85$.

Two caveats stated plainly. The normal approximation (A1) is the standard one for log-RR/OR/HR endpoints and is exact only asymptotically; it is least accurate at the smallest, lowest-base-rate cells, which is consistent with the abstention behaviour of §5. And because w is a property of the constraint polytope, these statements concern the input bands the polytope is built from: the runs in the body use $c = 1$ (exact marginally, conservative jointly), and any target joint coverage can be regenerated through the Bonferroni relation without re-solving the structure.

B. Monotonicity of the validation window

This appendix states the per-study coverage property formally and proves that the validation window is monotone under evidence addition: a consistent literature constraint can only narrow the window, a binding one strictly narrows it, and an inconsistent one inflates it. Two scalars set the per-study band: the Bonferroni multiplier $c \geq 1$ (fixed; $c = 1$ gives the marginal 95%, $c \approx 1.85$ gives joint 95% for our $n = 166$ studies; the body uses $c = 1$), and the binary-searched feasibility scalar $w > 0$ (the smallest scaling at which the constrained QP stays feasible; its per-edge polytope projection is the feasible range r). The argument below tracks how w moves as constraints are added at fixed c.

Setup. For a non-root node Y with parents $X_1, \ldots, X_K$, write $\mathbf{x} \in \mathbb{R}^M$ for the flattened CPT. A literature study i reports a point estimate s_i and 95% half-width CI_i for the marginal $P(Y{=}y \mid X_{p(i)}{=}v_i) = R_i^\top \mathbf{x}$, where R_i encodes the law-of-total-probability weights over the network's joint structure. The QP's per-study windowed band is $s_i - c\,w\,\mathrm{CI}_i \leq R_i^\top \mathbf{x} \leq s_i + c\,w\,\mathrm{CI}_i$. Cross-CPT subset constraints from the law of total probability take the form $S_j^\top \mathbf{x} = b_j$; CPT validity adds $\mathbf{x} \geq 0$ and the row-stochasticity equalities. The feasible polytope $\mathcal{P}(w, c)$ is their intersection, and the feasibility band of study i is its full projection width $r_i(w, c) = \max_{\mathbf{x}\in\mathcal{P}} R_i^\top \mathbf{x} - \min_{\mathbf{x}\in\mathcal{P}} R_i^\top \mathbf{x} \in [0, 1]$ (matching the window of §3). The solver determines $w^* = \min\{w : \mathcal{P}(w, c) \neq \emptyset\}$ by binary search and reports it as the window; the per-edge feasible range is $r_i^* = r_i(w^*, c)$.

Theorem 1 (Per-study coverage). Assume study i's literature 95% CI is well approximated by a normal interval at level $\alpha = 0.05$. Then for any $c \geq 1$ the band $s_i \pm c\,w^*\,\mathrm{CI}_i$ is a confidence interval for $R_i^\top \mathbf{x}^\star$ at level $1 - \alpha_c$, where $c\,z_{1-\alpha/2} = z_{1-\alpha_c/2}$ and z_q is the standard-normal q-quantile. In particular $c = 1$ recovers the marginal 95%, and $r_i^* \leq 2c\,w^*\,\mathrm{CI}_i$ (the full width is at most the band's, whose half-width is $cw^*\mathrm{CI}_i$).

Proof. Under the normal approximation the half-width CI_i equals $z_{1-\alpha/2}\,\mathrm{SE}_i$ for the study's standard error SE_i; multiplying by c widens it to $c\,z_{1-\alpha/2}\,\mathrm{SE}_i = z_{1-\alpha_c/2}\,\mathrm{SE}_i$, which is the stated relation. The bound $r_i^* \leq 2c\,w^*\,\mathrm{CI}_i$ holds because r_i is the full range of $R_i^\top \mathbf{x}$ over $\mathcal{P}(w^*, c)$, contained in the band of half-width $c\,w^*\,\mathrm{CI}_i$; the polytope is narrower than the band whenever other constraints (LoTP subset, simplex) are active on study i's axis. □

Convergence. The binary search on w is a one-dimensional bisection over a convex feasibility set, so w^* is unique given the constraints; across 10 independent initialisations w^* coincides to within solver tolerance (10^{-3} binary-search resolution, 10^{-6} in the underlying OSQP solve) on every per-cell QP.

Theorem 2 (Joint coverage). For n literature constraint rows tested at common nominal level $\alpha = 0.05$, the multiplier c giving joint 95% coverage by Bonferroni adjustment satisfies $c\, z_{1-\alpha/2} = z_{1-\alpha/(2n)}$. For the demonstration network ($n = 166$) this gives $c \approx 1.85$.

Proof. By the union bound, Pr(all bands cover their truth) $\geq 1 - n\,\alpha_c$ with α_c from Theorem 1; the bound holds under arbitrary dependence, and positive correlation among studies sharing reference cohorts (NHANES, UK Biobank) only makes coverage more conservative. Setting $1 - n\,\alpha_c = 0.95$ gives $\alpha_c = 0.05/n = 3.01\times10^{-4}$ for $n = 166$, so $z_{1-\alpha_c/2} \approx 3.61$ and $c \approx 3.61/1.96 \approx 1.85$. A Šidák adjustment is indistinguishable at this n. This is the band factor $c \approx 1.85$ of App. A; the body's runs use $c = 1$ (exact marginally, conservative jointly), and any target joint coverage follows from this relation without re-solving. □

Theorem 3 (Monotonicity of the validation window under evidence addition). Let $\mathcal{P}_n = \mathcal{P}(w_n^*, c)$ denote the feasible polytope after n literature constraints have been imposed at the binary-searched feasibility scalar w_n^*, and let $r_e^{(n)}$ denote the feasibility band of edge e at stage n (as in Theorem 1). Write $\mathcal{C}_{n+1} = \{x : s_{n+1} - c\,w_n^*\,\mathrm{CI}_{n+1} \leq R_{n+1}^\top x \leq s_{n+1} + c\,w_n^*\,\mathrm{CI}_{n+1}\}$ for the slab of the new study at scaling w_n^*.

1. (Weak monotonicity, consistent case.) If constraint n+1 is consistent with $\mathcal{P}_n$, i.e. $\mathcal{P}_n \cap \mathcal{C}_{n+1} \neq \emptyset$, then $w_{n+1}^* \leq w_n^*$, $\mathcal{P}_{n+1} \subseteq \mathcal{P}_n$, and consequently $r_e^{(n+1)} \leq r_e^{(n)}$ for every edge e.
2. (Strict monotonicity, binding consistent case.) If in addition $\mathcal{C}_{n+1}$ is binding on edge e (its bounding hyperplane properly separates at least one extremal point of $\mathcal{P}_n$ along the edge-e axis from the interior of $\mathcal{C}_{n+1}$), then $r_e^{(n+1)} < r_e^{(n)}$.
3. (Inflation under inconsistency.) If $\mathcal{P}_n \cap \mathcal{C}_{n+1} = \emptyset$ at scaling w_n^*, then $w_{n+1}^* > w_n^*$ and $r_e^{(n+1)} > r_e^{(n)}$ for at least one edge e touched by the new or an active prior constraint.

Proof of part 1. Fix the feasibility scalar at w_n^*. The feasible set at stage n+1 at this scalar is $\mathcal{P}_n \cap \mathcal{C}_{n+1}$, an intersection of convex sets ($\mathcal{P}_n$ by induction, $\mathcal{C}_{n+1}$ a single slab), nonempty by hypothesis, so the stage-(n+1) QP is feasible at w_n^*. By definition of w_{n+1}^* as the minimum feasibility scalar, $w_{n+1}^* \leq w_n^*$. Then $\mathcal{P}_{n+1} = \mathcal{P}(w_{n+1}^*, c) \subseteq \mathcal{P}_n \cap \mathcal{C}_{n+1} \subseteq \mathcal{P}_n$ (the first inclusion because a smaller w shrinks every slab). For any edge e, since $\mathcal{P}_{n+1} \subseteq \mathcal{P}_n$ the supremum of $R_e^\top x$ over the smaller set is no larger and the infimum no smaller, hence $r_e^{(n+1)} \leq r_e^{(n)}$. □

Proof of part 2. Let $x^* \in \arg\max_{x\in\mathcal{P}_n} R_e^\top x$, so $R_e^\top x^* = \sup_{x\in\mathcal{P}_n} R_e^\top x$. The binding hypothesis gives $x^* \notin \mathcal{C}_{n+1}$, hence $x^* \notin \mathcal{P}_{n+1}$. By part 1, $\mathcal{P}_{n+1} \subseteq \mathcal{P}_n$, so $\sup_{x\in\mathcal{P}_{n+1}} R_e^\top x$ is attained at some x^{**} with $R_e^\top x^{**} \leq R_e^\top x^*$. Because the separating hyperplane of $\mathcal{C}_{n+1}$ strictly separates x^* from $\mathcal{P}_{n+1}$ and (by the binding hypothesis) the separation direction has nonzero component on the edge-e axis, the inequality is strict, $\sup_{x\in\mathcal{P}_{n+1}} R_e^\top x < \sup_{x\in\mathcal{P}_n} R_e^\top x$. The same argument applies to the infimum if the new constraint also binds there; in either case the strict inequality propagates to the projection width, $r_e^{(n+1)} < r_e^{(n)}$. □

Proof of part 3. If $\mathcal{P}_n \cap \mathcal{C}_{n+1} = \emptyset$ at w_n^*, both sets are closed convex and a separating hyperplane exists. Each slab half-width is linear in w, so $\mathcal{C}_{n+1}(w)$ and every prior slab in $\mathcal{P}_n(w)$ expand with w, and joint feasibility $\mathcal{P}_n(w)\cap\mathcal{C}_{n+1}(w) \neq \emptyset$ becomes more permissive as w grows. By continuity of the boundaries in w there is a unique minimum $w_{n+1}^* > w_n^*$ restoring feasibility, at which at least one slab (the new constraint, or a prior constraint active at the old w_n^*) has its width strictly inflated, and the corresponding edge satisfies $r_e^{(n+1)} > r_e^{(n)}$. □

Remark (objective-invariance). Theorem 3 concerns only the constraint polytope. The QP's least-squares objective and monotonicity penalty (§2) select a single point inside $\mathcal{P}_n$ but do not enter the definition of r_e, so r_e is invariant to the objective weight λ, as stated in §3 and consistent with Theorem 1.

Remark (median). Part 1 gives $w_{n+1}^* \leq w_n^*$ and part 3 the strict increase under inconsistency; since the reported median $\tilde{w}$ is the median of the per-edge window values, it falls weakly as consistent evidence is added and strictly whenever a binding row lowers w^* across a positive fraction of edges. This is the formal underpinning of the median-window reduction reported in §4 (Table 1).

C. Determination of untested parent combinations: the LoTP activity threshold

Section 3, Property (4) asserts that the law-of-total-probability subset constraints determine untested parent combinations to within r, but only when the parent co-occurrence ρ is large enough to make the constraints tight. We make this precise.

Setup. Let Y be a binary node with K binary parents $X_1 \dots X_K$ and let $\mathbf{x} \in \mathbb{R}^{2^K}$ be its flattened CPT. Each marginal study i gives a linear equality $r_i^\top \mathbf{x} = s_i \pm \sigma_i$, where σ_i is the half-width of the literature 95% CI in CPT-cell units. Each subset constraint is a linear identity holding the law of total probability over a group of cells parametrised by the reference-survey co-occurrence ρ. Concretely, for two parents the law of total probability reads

$$P(Y \mid X_i{=}v_i) = P(Y \mid X_i{=}v_i, X_j{=}v_j)\, p + P(Y \mid X_i{=}v_i, X_j{\neq}v_j)\,(1-p),$$

with $p = P(X_j{=}v_j \mid X_i{=}v_i)$. Writing $m = P(Y \mid X_i{=}v_i)$ for the literature marginal and ranging the cofactor cell over its simplex bound $[0,1]$ pins the unknown joint cell to the interval $\big[\max(0, \frac{m-1+p}{p}),\ \min(1, \frac{m}{p})\big]$, of pre-clamp width $(1-p)/p$. Let σ denote the binding noise scale (the smallest σ_i relevant to the subset constraint).

The bite threshold (heuristic envelope). When $|\rho|$ exceeds

$$|\rho^\star| \;=\; \frac{\sigma}{1+\sigma}, \tag{2}$$

the subset constraint pins each untested cell to within a half-width $r \leq \frac{1-\rho}{1+\rho}\,\sigma$; below threshold it is non-binding and r relaxes to the marginal-only width σ. The body uses this $\sigma/(1+\sigma)$ form; it is the two-parent envelope of the rigorous threshold $\varphi(\sigma, K, p)$ derived and proved in the remainder of this appendix (Theorems 4 and 5).

Rigorous threshold. The heuristic (2) is the envelope of a rigorous threshold $\varphi(\sigma, K, p)$ that depends on the parent count K, the parent marginal p, and σ (the input-CI half-width as a fraction of the cell value, the QP's effective noise scale). We prove it under the equicorrelated Bahadur-1 multivariate Bernoulli model for the parent joint, with one literature row per parent; heterogeneous pairwise correlations are handled as a perturbation (Corollary 1), and higher-order parent-joint moments contribute a controlled error term discussed in the scope statement.

The Bahadur-1 ansatz. Under the equicorrelated Bahadur-1 model every parent has the same marginal p, every parent pair the same correlation ρ, and all higher-order central moments vanish. The joint takes the form $\pi_{\mathbf{v}} = \nu^{\text{ind}}_{\mathbf{v}} \big(1{+}\rho \sum_{i<j} \psi_{ij}(\mathbf{v})\big)$, where $\nu^{\text{ind}}_{\mathbf{v}} = p^{|\mathbf{v}|}(1{-}p)^{K-|\mathbf{v}|}$ is the independence joint, $z_k(v_k) = (v_k{-}p)/\sqrt{p(1-p)}$ is the standardized indicator, and $\psi_T(\mathbf{v}) = \prod_{k\in T} z_k(v_k)$ is the level-$|T|$ Bahadur basis function. The Bahadur basis is orthonormal under ν^{ind}; because Y's constraints lie in the level-≤ 2 subspace, the constraint-matrix analysis localises to a $\big(1 + K + \binom{K}{2}\big)$-dimensional subspace.

Network ρ vs. reference-survey ρ. Let $\hat{\boldsymbol{\pi}}$ be the parent joint identified by the QP from joint coverage in the reference population; the network pairwise correlation ρ^{net}_{ij} is its Bahadur level-2 coefficient. When joint coverage is sparse on a parent set, $\hat{\boldsymbol{\pi}}$ collapses onto the support of available joint observations and ρ^{net}_{ij} may differ from the marginal NHANES pairwise correlation $\rho^{\text{NHANES}}_{ij}$; the threshold below is stated in ρ^{net}, and for densely covered parent sets $\rho^{\text{net}} \approx \rho^{\text{NHANES}}$.

Theorem 4 (LoTP activity threshold, $K \geq 3$). Under the equicorrelated Bahadur-1 model with $K \geq 3$ binary parents (common marginal p, common pairwise correlation ρ^{net}), K literature rows (one per parent), per-output NHANES marginal, and at least one LoTP cross-CPT subset row S whose level-2 Bahadur projection has non-zero component orthogonal to $\text{span}(A^{(2)}_{\text{lit}})$, the LoTP constraint is active on at least one literature axis (i.e. $r_i^* < cw^*\text{CI}_i$ for at least one i) iff $\rho^{\text{net}} > \varphi(\sigma, K, p)$, where $\varphi(\sigma, K, p)$ is the unique root in $(0, 1)$ of

$$\rho \cdot \eta(K,p)\,(1-\theta_{\text{LoTP}}) = cws_{\max} \cdot \min\!\big((1-\rho)\,B_-(p),\ \rho\,\tau_2^+(K,p)\big). \tag{3}$$

The threshold admits a two-regime characterisation,

$$\varphi(\sigma, K, p) = \begin{cases} \dfrac{\sigma}{\sigma + \mu(K,p)}, & \sigma \le \sigma^\star(K,p) \quad \text{(level-1 binding)} \\ \dfrac{1}{1 + \sigma\,\mu'(K,p)}, & \sigma > \sigma^\star(K,p) \quad \text{(level-2 binding)} \end{cases}$$

with crossover $\sigma^\star(K,p) = B_-(p)/[cws_{\max}\tau_2^+(K,p)] \cdot \mu/(\mu+\mu')$. Here $B_-(p) = \sqrt{\min(p,1-p)/\max(p,1-p)}$ is the level-1 spectral floor, $\tau_2^+(K,p)$ is the smallest singular value of the level-2 sub-block of the constraint matrix after LoTP rows are added, $\eta(K,p)$ is the level-2 norm of the literature row R_i, and $\theta_{\mathrm{LoTP}} \in [0,1]$ is the fraction of R_i's level-2 signal absorbed by the LoTP row. The level-1 floor $B_-(p)$, the level-2 literature singular value $\tau_2(K,p)$ (4), and the row norm $\eta(K,p)$ are closed form (proof below); the only inputs fixed by the LoTP row's parent-subset geometry are τ_2^+ and the absorption fraction θ_{LoTP}, and across the operational range $K \in \{3,4,5,6\}$, $p \in [0.1, 0.5]$ these keep $\mu(K,p), \mu'(K,p) \in [0.5, 2.5]$.

Connection to the empirical envelope. The body operates at small literature-CI noise ($\sigma\approx$0.10–0.20), the level-1-binding regime, where $\varphi = \sigma/(\sigma+\mu)$ with $\mu \in [0.5, 2.5]$; the heuristic $\sigma/(1+\sigma)$ of (2) is the $\mu = 1$ case, correct as a multiplicative envelope of φ to within a factor of 2.5 across the operational range ($\mu \in [0.5, 2.5]$), which licenses its use in the body. (The level-2 form $1/(1+\sigma\mu')$ governs only the large-σ regime $\sigma > \sigma^\star$, outside the paper's operating point.)

Proof (equicorrelated Bahadur-1 ansatz). Step 1 (constraint matrix in the Bahadur basis). The literature row R_i clamping parent X_p at value a has Bahadur coefficients $\alpha_\emptyset = 1$, $\alpha_{\{p\}} = \beta_a$, $\alpha_{\{j\}} = \rho\beta_a$ for $j \ne p$, $\alpha_{\{i,j\}} = \rho$ for $i, j \ne p$, $\alpha_{\{p,j\}} = \rho\beta_a^2$ for $j \ne p$, and $\alpha_T = 0$ for $|T| \ge 3$, with $\beta_a = (a-p)/\sqrt{p(1-p)}$. The per-output marginal row m_Y has $\alpha_\emptyset = 1$, $\alpha_{\{i,j\}} = \rho$, else zero; the LoTP row S also has level cap ≤ 2. Step 2 (level-stratified $\sigma_{\min}$). The level-1 sub-block of A_{lit} has structure $\mathrm{diag}(\beta_{a_1}, \ldots, \beta_{a_K})\left((1-\rho)I_K + \rho\mathbf{1}\mathbf{1}^\top\right)$, whose smallest singular value is $(1-\rho)\,B_-(p)$. The level-2 sub-block is S_K-symmetric, so its Gram matrix carries only the trivial- and standard-representation eigenvalues; the standard-rep eigenvalue gives the closed form

$$\tau_2(K,p) = \sqrt{K-2}\,\left|\beta^2 - 1\right|, \qquad \beta^2 = \frac{(a-p)^2}{p(1-p)}, \tag{4}$$

so $\sigma_{\min}(A_{\mathrm{lit}}^{(2)}) = \rho\,\tau_2(K,p)$ and $\sigma_{\min}(A_{\mathrm{lit}}) = \min\left((1-\rho)B_-(p),\ \rho\,\tau_2(K,p)\right)$; we verified (4) against the network's constraint matrices for $K \in \{3, \ldots, 6\}$. Equation (4) vanishes at $K = 2$, where the two literature rows span only a rank-one level-2 block, so the literature cannot fix level-2 alone and the LoTP row must carry it: this is the separate base case of Theorem 5. It also vanishes at $p = \frac{1}{2}$, where the level-2 signal disappears. Step 3 (effect of the LoTP row). Adding S produces A_{full} with $\sigma_{\min}(A_{\mathrm{full}}^{(2)}) = \rho\,\tau_2^+(K,p)$, $\tau_2^+ > \tau_2$ (since S's level-2 projection has component orthogonal to $\mathrm{span}(A_{\mathrm{lit}}^{(2)})$). The polytope-projection bound (Vavasis and Ye, 1996; Dadush et al., 2024) gives $r_i \asymp \|P_{\ker A_{\mathrm{full}}} R_i\|/\sigma_{\min}(A_{\mathrm{full}})$. With $\|P_{\ker A_{\mathrm{full}}} R_i\| = \rho\,\eta(K,p)\,(1-\theta_{\mathrm{LoTP}})$ and the closed-form level-2 row norm $\eta(K,p) = \sqrt{\binom{K-1}{2} + (K-1)\beta^4}$ (verified likewise), the activity criterion $r_i \le cw\sigma s_i$ rearranges to (3), whose unique positive root is $\varphi(\sigma, K, p)$; the two-regime form follows from which term in $\min\left((1-\rho)B_-,\ \rho\tau_2^+\right)$ binds, with crossover at $\sigma^\star(K,p)$. □

Corollary 1 (Heterogeneous pairwise correlations). Drop equicorrelation: let ρ_{ij}^{net} have median ρ_{med} and spread $\Delta_\rho := \max_{ij,kl}|\rho_{ij}^{\mathrm{net}} - \rho_{kl}^{\mathrm{net}}|$. Provided the maximum sub-determinant of the parent-correlation matrix is bounded away from zero, the threshold $\rho_{\mathrm{med}} > \varphi(\sigma, K, p)$ continues to govern activity, with $|\text{exact threshold} - \varphi(\sigma,K,p)| \le C(K,p)\,\Delta_\rho$, where $C(K,p) = O(\sqrt{K}/\tau_2(K,p))$ is the matrix-perturbation Lipschitz constant of Tropp (2015, Cor. 6.1.6). This licenses using the median pairwise correlation as the operative quantity in §3 despite heterogeneous reference-survey correlation matrices.

Theorem 5 ($K = 2$ base case). Under the Bahadur-1 model with $K = 2$ binary parents, two literature rows, per-output NHANES marginal, and one LoTP cross-CPT subset row S with non-degenerate level-2 component, the LoTP constraint is active iff $\rho^{\mathrm{net}} > \sigma/(\sigma + \mu_2(p))$, where $\mu_2(p) \in [0.6, 1.4]$ for $p \in [0.1, 0.5]$. In our dataset $K = 2$ targets typically have one literature row per parent, so the 4-cell CPT is small-cell-determined by literature plus per-parent and per-output marginals plus simplex; the theorem is informative when fewer than two literature rows are available, and is included for derivational continuity with Theorem 4.

Scope and failure modes. Theorem 4 is rigorous under the equicorrelated Bahadur-1 ansatz. Five scope items. (i) Higher-order joint structure: the ansatz assumes parent-joint moments of order ≥ 3 vanish; real reference-survey joints have non-zero higher moments, introducing relative error $\sum_{k\geq 3} \rho_k^2/\rho_2^2$ (ρ_k the L^2-norm of the level-k block), empirically small on the parent sets we care about ($K \leq 5$). (ii) Heterogeneous marginals: equicorrelation assumes common p; heterogeneous p_k destroys the closed-form spectrum, and Corollary 1 covers heterogeneous correlations, not marginals. (iii) Missing literature on some parents: when the literature-row count is below K, the level-1 sub-block is rank-deficient and the level-2-binding regime governs. (iv) LoTP rows with degenerate level-2 component: a LoTP row from a downstream child sharing all K parents with Y can have its level-2 projection vanish, making LoTP structurally inert; the theorem applies to LoTP rows conditioning on a strict parent subset. (v) The constant θ_{LoTP}: the level-2 absorption fraction depends on the LoTP geometry, is generically $\Theta(1)$, and is treated as known by direct $\sigma_{\min}$ computation, absorbed into μ, μ'.

Related work on the threshold. The activity-threshold result sits at the intersection of three literatures. Credal networks and imprecise-probability propagation: Cozman (2000) characterises the strong-extension fixed point but gives no quantitative activity threshold; the CREMA library (Huber et al., 2020) implements constraint propagation as an algorithmic, not a constraint-matrix, property; the imprecise Dirichlet model (Walley, 1996) gives concentration intervals under i.i.d. multinomial data, not correlated parents. LP polytope geometry: the Vavasis–Ye $\bar{\chi}_A$-condition (Vavasis and Ye, 1996; Dadush et al., 2024) underlies Step 3's polytope-projection bound; the Restricted Isometry Property literature (Candès and Tao, 2005) preserves geometry under correlated random matrices but assumes isotropy/sub-Gaussianity, unlike our deterministic Bahadur structure. Constrained Bayesian-network learning: parameter-equality constraints (Niculescu et al., 2006), monotonicity constraints (Feelders and van der Gaag, 2006), and qualitative sign-propagation (Wellman, 1990) address hard constraints, not the soft CI-band activity our threshold characterises.

Practical reading. For typical literature CIs ($\sigma \approx 0.10$–0.20) the bite threshold is $\rho^\star \approx 0.09$–0.17. Parent pairs whose reference-survey co-occurrence reaches $|\rho| \geq 0.20$ are in the firm-bite regime: their combination CPT cells are determined to within r without any joint study. Pairs whose $|\rho| < 0.15$ are reported as wide-window w and surfaced on the widest-w list (Property 2): the system reports that it cannot extrapolate without joint data. Of the 336 parent-pair edges for which NHANES has ≥ 30 co-occurrence observations, 30% (102/336) reach $|\rho| \geq 0.15$ and 25% (84/336) reach $|\rho| \geq 0.20$; these drive their window to $w \leq 0.05$. The remainder cluster on the widest-w list and prompt either an additional connecting study or a structural change (a latent variable bridging the correlated co-parents). Median NHANES $|\rho|$ across all observable parent pairs is 0.044, so the bite regime is the upper tail of the distribution; but it is precisely those upper-tail edges that drive the window reduction of §4.

Two ways the window drives construction. The validation window improves the network through two levers. First, by ranking the worst-fit edges (the widest windows and largest objective-RR residuals), it nominates where a better study is needed or the surrounding structure must be repaired. Second, through the law-of-total-probability bite, it nominates co-parent pairs the reference survey correlates but the construction has left unconstrained. Applying the per-target threshold $\rho^\star = \sigma/(1+\sigma)$ to the canonical build: of the 123 multi-parent targets that carry both a comparable co-parent pair and a literature constraint, the reference-survey co-occurrence supports LoTP activity on about half (66), and the construction now realizes 34 of those, up from a third in earlier cycles as connecting edges were added (e.g. type-2-diabetes→osteoporosis). Closing the remaining gap (connecting the co-parents the population links but the construction has left at $\rho^{\mathrm{net}} \approx 0$, by an added connecting study or a bridging latent per the list above) is the operating loop's standing interconnection objective. This LoTP-realization gap and the window's abstain tier (App. M) are two views of the same residual under-determination, measured here by co-parent correlation, there by the per-edge window.

Connection to identifiability. The classical question of when marginal effects suffice to determine joint conditionals here receives a per-edge answer: the bite threshold (2) is the per-edge identifiability threshold, w is the per-edge non-identifiability radius, and the widest-w list is the sorted ranking of where in the DAG more evidence is needed. Identifiability is not a binary attribute of the model; it is a feasibility radius reported per edge and updated each cycle.

Empirical verification. Joining per-edge w with NHANES-measured $|\rho|$ on the network (194 edges) gives a step at the predicted threshold: median $w = 0.25$ for $|\rho| < 0.15$ (n=150, no bite); $w = 0.13$ for $[0.15, 0.20)$ (n=8, partial); $w = 0.001$ for $[0.20, 0.30)$ in QP-solved edges (n=19, full bite); the remaining 17 edges (to the 194 total) fall at $|\rho| \geq 0.30$ and are any_of aggregators or equivalent_to wrappers, whose wide w is aggregator saturation rather than a QP-solved edge and is correctly outside the theorem's domain. (These figures use the demonstration build of §4, where the windows retain the slack to resolve the step.)

D. Causal direction and confounding scope

The construction is causally directed. The DAG's edge directions are set by LLM reading of the agreed-upon causal mechanisms in the biomedical literature (review articles, mechanistic textbooks, and clinical-reasoning sources describing which exposures drive which outcomes through which intermediates). The LLM does not pick directions from association strength on the survey; it picks them from the literature's mechanistic consensus, the same channel a domain expert would use. Where that consensus is contested (depression $\leftrightarrow$ coronary artery disease), it takes the direction the majority of sources emphasise, and the residual disagreement surfaces later as a wide validation window or a direction-check failure. Direction-setting, edge extraction, and the structural placement of mediating nodes are the three things the LLM supplies that statistical analysis of the survey alone cannot. The result is a causal DAG that is encoded, not discovered: we run no causal-discovery algorithm over the data.

On confounding the construction is deliberately conservative. The network reported here fits the literature's observational RRs as published, without per-edge confounder adjustment. Two structural tools bear on confounding. dependency_distal wrappers (App. H) install a literature-backed correlation between two sibling parents the network would otherwise treat as independent (a shared-child v-structure), repairing false independence and countering aggregator washout while holding the disease's parent count fixed; this is a dependence-repair device, not a mediation decomposition, and true mediators are modelled separately as explicit $A \to M \to D$ chains. Separately, the software provides an optional, off-by-default deconfounding mode that, on the oriented DAG, converts a parent's total RR into a direct effect via the exact log-odds identity $\log \mathrm{OR}_{\mathrm{direct}} = \log \mathrm{OR}_{\mathrm{total}} - \log \mathrm{OR}_{\mathrm{indirect}}$; it is disabled in this build because subtracting modelled indirect effects from already-observational RRs risks double-adjustment.

What this licenses: the reported $P(Y \mid X = x)$ are literature-coherent observational conditionals organised by mechanistic causal direction, not identified interventional effects $P(Y \mid \mathrm{do}(X = x))$ in the sense of Pearl (1988). Meta-analytic adjustment sets across our source studies are heterogeneous and not tracked per row; the survey is cross-sectional and cannot separate forward from reverse causation on every edge; and where the mechanistic literature is wrong about a direction, the LLM inherits the error and the window cannot detect it. For deployment requiring interventional semantics, the additional no-unmeasured-confounding assumption is the deployer's to make and disclose. For this paper's claims (literature-coherent risk estimation, abstention where literature plus structure cannot commit, and research-direction signalling via widest-w edges), the observational conditionals are the right object, and the window reports the residual uncertainty per edge.

E. Effect-size harmonisation

The system accepts four input statistic types: relative risk (RR), odds ratio (OR), hazard ratio (HR), and standardised mean difference (SMD), normalising each to the marginal-RR form of Theorem 1 before entering the QP. Of these only the SMD is an effect size in the standardised-difference sense; RR, OR, and HR are ratio measures of association, and our network's inputs are predominantly RR.

OR $\to$ RR. Standard conversion $\mathrm{RR} = \mathrm{OR}/(1 - p_0 + p_0\,\mathrm{OR})$ using the target output's NHANES marginal p_0. For common outcomes ($p_0 \gtrsim 0.10$) this conversion can systematically overstate the effect, a known contributor to descriptive-tier residual error (App. M).

HR $\to$ RR. Hazard ratios are treated as time-stationary RRs without explicit conversion to a fixed-horizon cumulative incidence. HRs from studies with different follow-up windows are not directly comparable; the QP absorbs this incomparability into the residual rather than detecting it.

SMD → RR. Standardised mean differences are mapped to RR via the standard normal-link approximation with a guard against extreme values ($|\mathrm{SMD}| > 1$ flagged as a single-trial high-variance warning, blanked when the source is a single small RCT).

These harmonisations are sources of systematic bias the QP cannot detect from the constraints alone; they enter as bias rather than variance, and the feasibility band w does not bound them. The distinction matters for interpretation: w reports residual disagreement among the literature constraints as they reach the QP, not residual disagreement among the underlying epidemiological designs that produced them.

F. Study selection and quality

A single literature band enters the solver per (parent, value) cell: the constraint matrix carries one row per cell (§2), so two studies measuring the same cell are not stacked into two bands. Of the introduction's three epistemic challenges, cross-study disagreement on a single edge is therefore resolved upstream, at the extraction step (App. G), before the spreadsheet is built: when more than one study bears on a cell, the construction manual's selection criteria keep the single highest-quality study, while the polytope's job is the cross-edge joint coherence the law-of-total-probability subset constraints impose (App. C). Study-quality adjudication thus sits where the evidence is, leaving the polytope to address individual uncertainty and joint coherence.

Evidence hierarchy. Candidate studies for a cell are ranked meta-analysis > large population-based cohort ($n > 10{,}000$) > Mendelian randomisation or single cohort > single study, with pre-registered and peer-reviewed sources preferred; 72% of the demonstration network's rows are meta-analyses. Each surviving row records its sample size and study design (the spreadsheet's study_n and study_design fields), so the provenance of the selected study is auditable.

Inclusion checks. A row is admitted only if (i) its 95% CI excludes 1.0, so the study places a direction rather than straddling the null; (ii) it passes an optimal-information-size check, $n \geq 7.84\,[\,P_0(1-P_0)+P_1(1-P_1)\,]/(P_0-P_1)^2$ with $P_1 = P_0\,\mathrm{RR}$, so a small effect is not read off an underpowered sample; and (iii) it is drawn from a general population rather than a disease registry or a case-control sample of the target node, which would induce collider bias. When the study population does not match the reference survey's demographics, the effect size is population-corrected through the same NHANES baseline used in the OR→RR conversion (App. E), or, where transportability is doubtful, the study is treated as a direction indicator and magnitude upper bound rather than an exact target. Any adjustment to a reported statistic is recorded with its original value, adjusted value, and reason, so a corrected row stays reproducible.

Relation to the window. Selection fixes which study constrains a cell; the window then reports how much epistemic slack remains around that study once the joint constraints are imposed. A precise, well-powered, population-matched study yields a tight band; the cross-edge constraints may collapse its window below the literature CI (when they reinforce it) or leave it above (when the solver must stretch the study to stay feasible); a study admitted with a wide CI passes a correspondingly diffuse band downstream. The two mechanisms are complementary: selection controls per-edge input quality, the polytope controls joint consistency, and the window is the residual the second leaves after the first.

G. LLM-extraction validation

Because the spreadsheet rows are produced by an LLM reading published studies, extraction error is a primary risk, and three controls bound it. (1) Blind re-extraction. A second LLM agent, given a differently phrased prompt, independently re-extracts the effect size, CI, population, and PMID for each edge; rows where the two agents disagree on the effect size beyond a tolerance, or name different source papers, are held for human adjudication rather than admitted. (2) Citation verification. Every cited PMID is resolved against PubMed and its title and abstract are checked for topical match to the encoded relationship; rows whose PMID does not resolve, or resolves to an unrelated paper, are dropped. An audit of the LLM-extracted corpus under this procedure found and corrected a minority of rows carrying mis-resolved citations or out-of-tolerance effect sizes. (3) Two solver-side catches that need no external ground truth. Each edge's encoded direction is checked against the network's query-RR, so a sign flip is surfaced (§4); and the law-of-total-probability subset constraints reject any set of cells that cannot be made jointly coherent. A mis-transcribed effect size therefore appears either as a direction-check failure or as a widened validation window, so the extraction step needs only to be auditable, and w is the audit channel.

Audit results. We hand-validated a random sample of 30 study rows drawn uniformly without replacement (random.seed = 7) along five dimensions: paper existence, title match, journal match, year match, and semantic correspondence between the CSV's input variable and the paper's exposure. Of the 30, 29 resolved to real, verifiable PubMed-indexed papers; one had no citation and was removed pending re-extraction. Five rows carried non-substantive metadata noise (journal-field errors, year off by one or three, paraphrased rather than verbatim titles, one design-code mismatch), corrected without affecting the QP, which consumes only the statistic, its CI, and the parent-child structure. One row applied a smokeless-tobacco effect estimate to an input variable labelled ever_vaped (pharmacologically distinct exposures sharing the surface form "tobacco-adjacent"), and was corrected. A second random sample (random.seed = 11, no overlap) gave 30/30 bibliographic accuracy and 28/30 semantic match. Combined bibliographic accuracy across both samples is 59/60 = 98.3% (95% Clopper–Pearson $[91.1\%, 99.96\%]$); semantic match was 28/30 on the second sample, with the first sample reported below under the more conservative independent re-extraction (we do not pool the two semantic counts, which use different protocols).

Self-critical finding from independent re-extraction. An independent re-extraction of the same n=30 first sample by a separate LLM identified four substantive errors the single-pass audit missed: one wrong-paper attachment, two U-shape papers where the LLM took one tail's HR for the opposite-tail row, and one paraphrased-vs-mismatched title at PMID 27744349. This puts the substantive-fidelity rate at 26/30 on the first sample under independent re-extraction, below the 29/30 of the single pass; we do not claim the remaining unchecked rows are error-free at the higher rate. The window w and the per-row direction test are the operational catches for errors of this kind: the architectural assumption is that the LLM needs to be auditable rather than perfect, with w as the audit channel.

H. The BayesExpert construction vocabulary

The Type column on each row declares either a node type (a node's role and CPT) or a link type (a definitional or routing relationship between two nodes). We summarise the vocabulary; the construction manual released with the code gives the full grammar.

Node types (declared on a definition row).

- NHANES leaf: a directly-observed survey variable. A naive_0 leaf has its CPT computed from NHANES joint counts against a parent, subject to guards against degenerate or definitionally-circular joints.
- dependency: a node whose CPT $P(Y \mid X_1, \ldots, X_K)$ the QP solves from the study RRs on its parent edges (the default study-backed node), with variants for NHANES-coded and prior-only parents.
- any_of / all_of: logical aggregators whose CPT is a deterministic OR or AND truth table over their parents, collapsing many correlated leaves into a single binary node instead of K separate parents.
- dependency_distal: the wrapper node described below.

Link types (a relation between two nodes).

- is_a / subsumes: definitional containment, $A \subset B$ encoded by sensitivity = $P(A)/P(B)$, specificity = 1 (so $P(B \mid A) = 1$), and its inverse.
- equivalent_to / equivalent_distal: identity, sensitivity = specificity = 1 (so $P(A) = P(B)$); the channel by which an NHANES measurement reaches a canonical node, with equivalent_distal the identity link used inside a distal wrapper.
- distal: a routing link whose aggregator-to-disease RR is not a study but is computed by aggregating the aggregator's leaf RRs weighted by NHANES co-occurrence (a Monte-Carlo over the leaf CIs gives the disease-edge CI). A leaf row tagged with a disease name, e.g. output=chemical_pc_risk, input=nickel_exposure, Type=pancreatic_cancer, RR=1.7, routes that leaf's RR (1.7) to the named disease through such a chain.

The dependency_distal wrapper. The distinctive construct is a three-row block that adds a connecting-study correlation between an aggregator's output and a correlated sibling node without raising the parent count K at the downstream disease. The block is: (i) an equivalent_distal identity row routing the original aggregator into the wrapper unchanged (sensitivity = specificity = 1); (ii) the dependency_distal definition row declaring the wrapper node and its states; (iii) a regular literature-RR row whose input is the correlated sibling, pinning a marginal of the wrapper's CPT to a literature-anchored value that propagates into the disease's marginal through the law of total probability. The wrapper treats two regimes the plain construction handles poorly: under-determined interaction cells at multi-parent targets whose parent-pair correlation is below the LoTP-activity threshold, and signal attenuation in long aggregator chains (each wrapper layer roughly halves the queried effect, App. I), where anchoring the wrapper's CPT against a sibling restores literature-shape behaviour however attenuated the upstream signal. The disease still sees a single distal-typed parent, so the cross-edge correlation is added without the 2^K cost of an extra direct parent.

I. Wiring rules and structural checks

The construction guidance of §6 reduces to wiring rules that keep the polytope informative as edges are added, each preventing a specific saturation or inversion failure mode (App. H); the rules are enforced by structural data-checks run before every build.

Four wiring rules (logical gates). (R1) Trigger each leaf on its least-prevalent state, with the RR direction set accordingly: rare-side triggers carry information, same-side triggers saturate the gate. (R2) Limit K per gate. An any_of gate collapses K inputs to one yes/no, losing more information at higher K; target $K = 2$–3, hard ceiling $K = 7$. (R3) Do not group correlated leaves into one gate. Leaves with $|\rho| \geq 0.30$ saturate the OR-mask and lose the cross-leaf evidence flow the QP would otherwise use; cluster them under a common-cause latent or a multi-rung wrapper ladder (chain-depth note below). (R4) Do not mix opposite-direction RRs in one gate. Risk inputs (RR > 1) and protective inputs (RR < 1) routed to the same any_of cancel via the QP fit, producing direction-inverted residual signal; split them into separate sub-gates.

Five within-aggregator sibling-discipline rules. Within one aggregator or wrapper, all sibling rows must satisfy: (S1) same direction on the disease (mixed signs cancel); (S2) near-magnitude RRs (strong siblings dominate the QP fit; empirical envelope max / min ≤ 3); (S3) low-prevalence triggers (states with marginal $P > 0.30$ saturate the OR-mask); (S4) effect traced empirically (set evidence on the leaf, query up the chain, confirm the direction preserves to the disease); (S5) semantic coherence (siblings represent the same kind of thing the gate names; a physical-activity sibling does not belong inside a healthy-diet aggregator even though both predict diabetes, since semantic confusion produces phantom signal flows that double-count when the node also feeds its proper gate).

Structural data-checks. The grammar is enforced by automated checks (scripts/data_checks.py) before every build. Wrapper value-name alignment verifies that every dependency_distal wrapper's exposed-state columns name states that exist as outvars on the wrapper's primary-chain parent (a misalignment would otherwise silently drop the consumer's evidence at inference); consumer state-name reachability verifies that every consumer row's input-value reference matches one of the wrapper's exposed states. Other checks enforce $K \leq 7$, no self-reference, no orphan inputs or outputs, no row missing input values, and no duplicate (output, input, input-value) tuples. An LLM that emits a violating row sees the failure message and revises before the QP solver is invoked.

Chain depth trades magnitude for K. Empirically each wrapper layer between a leaf and the disease attenuates the leaf's queried effect by roughly half, so depth trades magnitude for the K-reduction it buys; a leaf that must retain its magnitude is promoted to a direct disease parent.

J. Per-respondent ROC AUC and discrimination metrics

The metrics below are computed on our network (573 nodes: 84 directly-observed NHANES variables, 26 definitional sub-variables, 307 literature-solved dependency nodes of which 15 are disease targets, 149 logical aggregators, and 7 distal aggregators), the build whose median validation window is 0.001. Direction-accuracy and window figures use the corrected per-edge reference-category metric; the residual direction inversions concentrate in the low-base-rate cancer chains that §5 characterises as abstentions.

What each measure is. Every figure in this appendix is computed on held-out NHANES respondents or on the solved network directly. Per-respondent ROC AUC: roc_auc_score of the network's posterior $P(\text{target} \mid \text{evidence})$ against the held-out binary outcome, over n=300 respondents per target, with each target's definitional surrogates removed from the evidence (App. L). Objective-RR direction and within-$X\%$: for each study edge the network is queried with the parent at the study's exposure value, giving a query-RR q_{rr} (exposed vs. unexposed reference group); direction is the sign of $q_{\text{rr}}-1$ checked against the study's, and "within $X\%$" means relative error $|q_{\text{rr}}-\text{RR}|/\text{RR} \leq X$. Log-ratio error: $|\ln(q_{\text{rr}}/\text{RR})|$, a symmetric multiplicative discrepancy that a sign flip drives large. Validation window w: the per-edge feasibility scaling of §3 (the smallest w keeping the cell's QP feasible), summarised as the median and mean over the solved per-cell CPT rows. Reliability ECE / MCE: predictions are sorted into ten equal-width probability bins; ECE is the count-weighted mean of $|\overline{p}_{\text{bin}} - \text{frequency}_{\text{bin}}|$ and MCE its maximum over bins. Brier score: the mean squared error $\frac{1}{n}\sum_i(p_i - y_i)^2$ between predicted probability and outcome. Sharpness: the variance of the predictions, $\text{Var}(p)$, which a base-rate-only predictor drives to 0; a useful net has low ECE and non-trivial sharpness. Mean $|\rho\text{-gap}|$: over comparable parent–co-parent pairs, the mean absolute difference between the correlation the network induces and the same correlation measured in NHANES, an internal check that the solved CPTs reproduce the reference survey's joint structure. Under-determined nodes: those whose mean window exceeds the identifiability threshold 0.05, i.e. that the current literature does not pin down. The chain measure of Table 5 and the UK Biobank quantities are defined where they appear.

Per-target discrimination. On held-out NHANES respondents, with each target's own NHANES code and its definitional-surrogate biomarkers excluded from the evidence set (so the model cannot read the answer off a defining measurement, App. L), per-respondent ROC AUC averages 0.775 across the 12 scored disease targets (12 of the network's 15 disease targets; three non-population nodes held out) (range 0.571–0.945). Several targets reach or exceed the level of cohort-fitted clinical risk calculators (≈0.71 for Framingham 10-year CVD and pooled ASCVD; a level comparison across cohorts, not a head-to-head run on the same respondents): coronary artery disease 0.95, depression 0.89, diabetes 0.85, insomnia 0.85, sleep apnea 0.81, stroke 0.76, heart attack 0.76, COPD 0.74, and cognitive impairment 0.73; the remainder fall below calculator level, down to cancer at 0.57. Per-target AUC tracks whether the network's literature supplied informative risk-factor parents for the target. Table 2 gives the full breakdown.

Target	AUC	Prevalence
Coronary artery disease	0.95	0.057
Depression	0.89	0.063
Diabetes	0.85	0.07
Insomnia	0.85	0.16
Sleep apnea	0.81	0.037
Stroke	0.76	0.03
Heart attack	0.76	0.043
COPD	0.74	0.053
Cognitive impairment	0.73	0.16
Hypertension	0.70	0.28
Osteoporosis	0.69	0.07
Cancer	0.57	0.097
Mean	0.775	

Table 2. Per-respondent ROC AUC by target on the network, with definitional-surrogate biomarkers excluded per App. L. Scored with roc_auc_score on held-out NHANES respondents (n=300 each). AUC tracks the informativeness of each target's literature-supplied parents, not a ranking of the diseases.

Objective-RR fidelity. Beyond direction, we score the network's query-RR against the literature RR on every edge that carries one: the solved network is queried with the parent clamped to the study's exposure value, and the resulting RR (exposed vs. unexposed reference group, App. L) is compared to the cited point estimate. The 445 testable edges exceed the 193 studies of the 573-node build because one study typically constrains several parent–outcome edges. Each cited edge falls into one of three directional outcomes that partition the panel (Table 3): the network reproduces the study's direction on 380 (85.4%), washes out on 56 (12.6%, the study is directional but the queried network does not move), and reverses the study's sign on 9 (2.0%). The washed-out edges are the aggregator-chain attenuation of §5: their fraction rises with chain depth (Table 5) as the law-of-total-probability signal weakens across more intervening nodes, and the network abstains rather than commit to a direction. On the edges it reproduces, the queried magnitude is close on the multiplicative scale risk ratios live on: the median discrepancy is a factor of 1.16 (a study reporting a 5× risk against the network's ≈4–6×), and 90% of all edges land within a factor of 2 of the cited RR (median absolute log-ratio 0.19). The mean log-ratio (0.33) is inflated by a handful of low-base-rate cancer chains whose queried RR is extreme at near-zero base rates (the abstention regime again); the median is the outlier-resistant summary.

Network query-RR vs. literature RR (445 edges)	Value
Direction reproduced	380 (85.4%)
Washed out (directional study, network unmoved)	56 (12.6%)
Reversed	9 (2.0%)
Median discrepancy factor, reproduced edges	1.16×
Within a factor of 2 of the literature RR	89.8%
Within 50% of the literature RR	85.2%
Median absolute log-ratio $\lvert \ln(q_{\mathrm{rr}}/\mathrm{RR}) \rvert$, all edges	0.19
Mean absolute log-ratio (low-base-rate tail)	0.33

Table 3. Objective-RR fidelity: how each edge's queried RR reproduces the literature RR it was built from. The three directional rows partition the 445 cited edges: reproduced (the network moves the disease the study's way), washed out (directional study, unmoved network query, the aggregator-chain attenuation of §5), and reversed (opposite sign). Magnitude is on the multiplicative scale risk ratios live on (a factor, $e^{\lvert \ln(q_{\mathrm{rr}}/\mathrm{RR}) \rvert}$); the per-edge RR uses the exposed-vs-unexposed reference category, not the population marginal.

Complete measurement panel. Table 4 collects the remaining build-level metrics. The median validation window is 0.001 (mean 0.086); reliability ECE is 0.014 and Brier 0.070; the maximum single-bin calibration error (MCE) is 0.32, confined to a sparsely-populated bin so the count-weighted ECE is the representative summary; prediction sharpness (variance) is 0.021; the mean gap between the network's induced parent–co-parent correlation and the NHANES correlation is 0.091. Sex-stratified AUC is 0.780 (male) versus 0.776 (female) (by-sex reliability ECE 0.014 male, 0.016 female); we report the gap as a monitoring target without attributing a cause. Of the 370 internal nodes carrying a solver-computed validation window (the other 203 of the 573 are the 84 directly-observed inputs and 119 deterministic or unsolved nodes, which carry no window), 83 remain under-determined by the current literature; their predictions revert to the solver's prior, where the window reports the residual uncertainty.

Metric	Value
Median validation window $\tilde{w}$	0.001
Mean validation window	0.086
Per-cell rows with $w > 0.05$	222 / 1029
Reliability ECE / MCE	0.014 / 0.32
Brier score	0.070
Prediction sharpness (variance)	0.021
Mean $\lvert\rho$-gap$\rvert$ (network vs. NHANES)	0.091
ROC AUC, male / female	0.780 / 0.776
Reliability ECE, male / female	0.014 / 0.016
Under-determined internal nodes	83 / 370

Table 4. Complete build-level measurement panel on the network (573 nodes), supplementing the per-target AUC of Table 2 and the objective-RR fidelity of Table 3. Reliability ECE/MCE, Brier, and sharpness are pooled over all held-out scored respondents; the validation-window rows are over the 1029 solved per-cell CPT rows.

Signal across chain length. A stricter measure asks whether a study's directional signal is retained along the full path from leaf to disease, scoring any edge that flattens (washes out to no movement) or reverses the literature direction as a failure. Resolved by the number of hops from evidence to target (Table 5), retention declines with depth, from 204/230 at one hop to 4/10 at five (379/435, 87% overall). It is in line with the panel's 85.4% per-edge figure (here restricted to the 435 edges on a multi-hop leaf-to-disease path), since both score a flat edge as a failure; the flattened fraction rises with depth as the law-of-total-probability signal weakens across more intervening nodes, which is exactly the aggregator-chain washout of §5. (The 33 flattened here and the 56 washed-out in Table 3 are different measures and do not subtract: this table scores directional retention along each leaf-to-disease path, resolved by hop count, whereas Table 3 scores each cited edge's own query at its target; the 435 hop-resolved paths are not a subset of the 445 cited edges.)

Hops (leaf→target)	Edges	Direction retained	Reversed	Flattened	Rate
1	230	204	16	10	88.7%
2	122	116	3	3	95.1%
3	43	37	4	2	86.0%
4	30	18	0	12	60.0%
5	10	4	0	6	40.0%
Total	435	379	23	33	87.1%

Table 5. Directional-signal retention by chain depth. Each leaf-to-disease edge is retained (literature sign kept), reversed (non-flat but off the literature sign), or flattened (washed out to no movement), and the three columns sum to the edge count; Rate = retained/edges. Reversals concentrate at shallow hops, while flattening dominates the depth-4 and depth-5 rows (the long any_of chains of §5). The 23 reversed and 33 flattened are this per-hop-path measure and need not equal Table 3's 9 per-edge sign reversals and 56 wash-outs, which score each cited edge's own query.

Why the metrics are complementary. Each measure answers a different question and is blind where another sees. The validation window w is local: given the network's parent co-occurrence, it reports how tightly the literature and the law of total probability pin one edge's cell. It conditions on that co-occurrence rather than checking it, so a network whose induced parent correlations are wrong but internally consistent still yields tight windows; the ρ-gap audit (Table 4) is the separate check that the network's parent correlations match the population's. The window reports the radius of the feasible region, not its symmetry, so it passes over the low-base-rate edges whose region admits a direction-inverted point (§5); the objective-RR direction check (Table 3) catches those. Discrimination and calibration in turn measure predictive use, which neither addresses. Internal consistency is not identical to correctness, but the two are not independent: because the checks overlap, through cells shared by the law of total probability, the common population marginals, and the literature on adjacent edges, an isolated wrong value cannot stay hidden, since it would have to be fortuitously consistent with every constraint touching it at once, which grows less likely as the network densifies. Broad coherence across a densely cross-constrained network is therefore evidence of correctness, not merely of internal tidiness. What it does not defeat is bias shared systematically across the literature, consistent by common cause rather than coincidence; that residual is what the external UK Biobank comparison (App. K) is for, the one test that brings in data the construction never saw.

K. UK Biobank external validation

Broad direction sweep (n=50). We compare the network's query-RRs, read live from the canonical build, against published UK Biobank cohort effect sizes on 50 matched exposure-outcome pairs not used in construction. 43/50 (86%) agree in direction with the UK Biobank estimate, and 42/50 (84%) fall within 50% of it. This is a held-out comparison against a cohort distinct from both the construction literature and the NHANES reference set.

Tier-spanning three-way comparison (n=20). On a smaller subset spanning all three tiers we additionally compare each net query-RR against both the UK Biobank estimate and the raw NHANES parent–target marginal RR (Table 6), which isolates where the network's prediction comes from: when the NHANES marginal disagrees with the literature, does the net follow the literature it was built from, or does it merely echo the population correlation? Across the 20 pairs the net agrees with the UK Biobank direction on 18 and reverses it on two (high-milk→fracture and low-fibre→fatal-MI); on three of the 18 the net abstains (query-RR ≈1) rather than commit to a signed effect (the cohort estimate is protective on the two physical-activity rows and a 1.50 risk

#	Tier	Target ← exposure	Lit RR	Net qRR	UK Biobank	NHANES marg.
1	tight	hypertension ← insomnia / short sleep	1.21	1.20	1.20	2.83
2	tight	fracture ← high milk intake	0.57	0.63	1.09	n/a
3	descr.	prostate cancer ← IGF-1 (Q4)	1.29	1.31	1.25	n/a
4	tight	metabolic syndrome ← current smoking	1.26	1.26	1.73	n/a
5	descr.	all-cause mortality ← frailty	2.00	1.12	2.70	n/a
6	descr.	all-cause mortality ← long sleep	1.34	1.17	1.16	n/a
7	descr.	all-cause mortality ← vitamin-D defic.	1.57	1.12	1.20	n/a
8	tight	hypertension ← obesity (BMI)	1.71	2.38	1.64	2.97
9	descr.	diabetes ← obesity (BMI)	4.56	1.68	5.50	3.68
10	abstain	lung cancer ← current smoking	8.43	2.86	8.00	2.99
11	tight	hypertension ← sleep apnea	1.53	1.23	1.56	2.42
12	descr.	all-cause mortality ← obesity (BMI)	1.28	1.16	1.21	n/a
13	descr.	diabetes ← high activity (Q4)	0.65	1.00	0.51	0.99
14	descr.	diabetes ← moderate sport (Q4)	0.68	1.00	0.51	0.99
15	descr.	breast cancer ← severe obesity (BMI)	1.57	1.25	1.65	0.00
16	tight	fatal MI ← low fibre (Q1)	1.20	0.67	1.13	1.07
17	tight	metabolic syndrome ← early menarche	1.62	1.62	1.62	n/a
18	tight	fatal MI ← snus	1.13	1.32	1.37	2.32
19	tight	heart disease ← snus	1.17	1.15	1.37	2.05
20	tight	fracture ← vegetarian diet	0.80	1.00	1.50	n/a

Table 6. Tier-spanning three-way external comparison on the canonical build: the net's queried RR against the UK Biobank cohort estimate and against the raw NHANES parent–target marginal RR, the latter shown only for the eleven targets that carry a NHANES code ("n/a" otherwise). Net qRRs are queried live from the build pickle (exposed vs. unexposed reference category). Direction-concordant with UK Biobank on 18/20; within 50% of the UK Biobank RR on 15/20. Rows 1 and 15 are the clearest cases where the NHANES marginal disagrees with the literature and the net follows the literature onto the UK Biobank value; rows 18–19 (smokeless tobacco) are further such cases, where a thin NHANES cell over-states the marginal while the net tracks the literature.

on vegetarian-diet→fracture), and it is within 50% of the UK Biobank RR (relative error ≤ 0.5) on 15. The decisive cases are the two whose target carries a NHANES code on which the NHANES marginal is misleading: for insomnia→hypertension the raw NHANES marginal RR is 2.83, but the literature (1.21), the net (1.20), and UK Biobank (1.20) all agree near 1.2; for severe-obesity→breast-cancer the NHANES co-occurrence is degenerate at 0, yet the net follows the literature (1.57) to 1.25, close to UK Biobank's 1.65. Where the population marginal would mislead, the network tracks the literature and still generalises to a third cohort, which is the external signal that the construction adds information beyond the NHANES correlations rather than re-encoding them.

L. Per-target diagnostic-biomarker exclusion

The per-respondent AUC of App. J is meaningful only if the model cannot read a target's answer off a measurement that defines that target. We therefore exclude, per target T, every evidence node X that is a definitional surrogate of T, by a strict rule: X is dropped from the evidence for T iff it is (a) a deterministic aggregator ancestor of T; (b) a same-NHANES-code alias of T; (c) a diagnostic biomarker that defines the disease threshold, e.g. HbA1c, fasting glucose, and HOMA-IR for diabetes, the last excluded because it is a deterministic function of the already-excluded fasting glucose and insulin; (d) the NHANES screening item that is the disease question; or (e) for an umbrella diagnosis, a subtype reachable through the umbrella's membership channel (the specific cancers under an all-cancer target; emphysema and chronic bronchitis under COPD). The rule deliberately does not exclude risk factors, sibling diseases, downstream mortality, or merely-correlated clinical observations: these are legitimate evidence. Symptom items with diverse non-disease causes (individual depression-screen items, single-night sleep complaints) are likewise retained: predictive but not definitional. This separation is what distinguishes a discrimination estimate from one inflated by a biomarker that is the diagnosis under another name; without it the apparent AUC on several targets rises sharply but reports only definitional equivalence, not prediction. A second, weaker channel is the construction's population statistics: the NHANES base rates and parent co-occurrence frequencies the solver fits are computed over all respondents, the held-out evaluation split included. These enter only as marginals and aggregate joint frequencies, never as a per-respondent conditional $P(\text{target} \mid \text{evidence})$ carrying a held-out outcome, and the target base rate in particular is a single per-target constant that shifts every posterior by the same prior factor, moving the operating point rather than the rank order the AUC reads; the per-respondent ranking therefore reflects the solved network's structure, not memorized held-out outcomes. Table 7 gives the complete per-target exclusion list applied before scoring.

Target	Excluded definitional surrogates
Coronary artery disease	angina, heart attack, the heart-attack screen alias
Heart attack	angina, coronary artery disease, the heart-attack alias
Diabetes	HbA1c, fasting glucose, random serum glucose, HOMA-IR, insulin (two codings)
Hypertension	systolic, diastolic, blood-pressure-medication prescription and compliance
COPD	emphysema, chronic bronchitis, the COPD screen alias
Cognitive impairment	dementia, Alzheimer's, non-Alzheimer's dementia
Cancer (umbrella)	the site-specific cancers reachable through the all-cancer membership channel: breast, colon, gallbladder, kidney, laryngeal, leukemia, liver, lung, melanoma, oesophageal, oral, ovarian, pancreatic, prostate, non-melanoma skin, stomach, thyroid
Sleep apnea	the sleep-apnea screen alias, snoring frequency
Depression	the PHQ-9 functional-impairment add-on item
Insomnia	the insomnia screen alias
Stroke	fatal stroke
Osteoporosis	none

Table 7. The complete per-target definitional-surrogate exclusion list, generated against the build's configuration and applied before scoring the AUC of Table 2. Each excluded node has a NHANES code and either is a deterministic alias of the target, a diagnostic biomarker that defines its threshold, the screening item that is the disease question, or (for the cancer umbrella) a subtype reachable through the membership channel. Risk factors, sibling diseases, downstream mortality, and merely-correlated observations are not excluded.

M. Confidence tiers induced by the window

The validation window induces a natural confidence tiering of the network's edges, and we find the tier predicts external generalisation. An edge (target Y) is tight-committed when its CPT is effectively determined: either small enough that literature, population marginals, and the simplex bounds supply as many equations as unknowns, or with a parent pair correlated strongly enough that the law of total probability is active ($|\rho| > \sigma/(1+\sigma)$). It is descriptive when direction and rough magnitude are pinned but the window is wider, and abstain when the polytope is symmetric along the direction axis (§5). This window tier (a property of the edge's whole polytope) is distinct from a single clamped query abstaining numerically at $q_{\mathrm{rr}}\approx 1$, which a tight- or descriptive-tier edge can also do on a particular parent value (the UK Biobank rows of App. K). The external UK Biobank checks (App. K) are consistent with this reading at the aggregate level (43/50 on the broad sweep and 18/20 on the tier-spanning three-way of Table 6); both external samples remain too small to resolve a separate per-tier generalisation rate (on the 20-pair set the tight, descriptive, and abstain tiers run 8/10, 9/9, and 1/1), which we leave to a larger external cohort. The practical use is triage: a deployment trusts tight-committed edges, treats descriptive edges as directional only, and routes abstain edges to the window report rather than a point answer.

Tier distribution. Across the 122 scored nodes (distinct from the 12 scored disease targets of App. J) that carry objective-RR comparison rows on the canonical build, the window sorts 27 (22.1%) as tight-committed, 52 (42.6%) as descriptive, and 43 (35.2%) as abstain. The abstain tier is dominated not by direction failures (18 targets) but by magnitude-undetermined low-base-rate chains (25 targets), the cancer-risk edges of §5 whose queried RR the solver cannot pin at near-zero base rates, where the system correctly routes to the window rather than a point estimate.

N. Crowdsourced construction and continual improvement

The construction is designed to ingest literature continuously rather than as a one-time corpus. New study rows are proposed by the LLM extractor (App. G) and may also be contributed by domain experts through a public crowdsourcing protocol (CROWDSOURCING.md); each proposed row is admitted only after the two-agent quorum and PubMed verification of App. G and review by a human curator, who sets the construction direction, adjudicates the rows the automated checks flag, and decides what enters the released build. Admitted rows are not averaged into a fixed network; each is an update operator that re-solves the quadratic program against the existing polytope. A row consistent with the rest of the network shrinks the polytope and lowers the median window; a conflicting row forces the feasibility factor w wider and surfaces in the widest-window list (§4) together with the specific edges it disagrees with. This yields a principled acceptance criterion for crowdsourced evidence (accept if the median window decreases, flag for adjudication if it increases, returning edge-specific feedback) and a continual-improvement loop whose running record of accepted changes and their effect on the median window is public (IMPROVEMENT_LOG.md): the collective polytope width is a monotone-decreasing measure of residual epistemic uncertainty that the system drives down both by adding consistent literature and by the structural changes its widest windows nominate. Because every update re-solves rather than overwrites, the window remains an accurate report of what is unresolved at each point in the network's life, which a lifelong-learning deployment requires. Where gradient-based continual learning must regularise against catastrophic forgetting (Kirkpatrick et al., 2017), the re-solve here has no weights to overwrite: a consistent row only shrinks the polytope, so prior evidence is retained by construction rather than by a penalty term.

The construction as a logged optimization. In practice the network is not built in one pass but optimised over many cycles, and the quantity driven down is the median validation window itself, the same $\tilde{w}$ the system reports, subject to the legitimacy gates every build must pass (acyclic DAG, citation quorum, no degenerate conditional tables). Each cycle applies one update, an added study row or a structural change, re-solves, and records the resulting window distribution, so the construction leaves a continual-improvement log whose accepted entries form the monotone-decreasing trajectory of $\tilde{w}$. Table 1 is an excerpt of that log: the within-baseline rows trace $\tilde{w}$ from 0.125 to 0.016 as consistent literature edges are added, and the post-rebuild row reaches 0.001 after the structural revisions the widest windows themselves nominated. Direction accuracy and joint fidelity ride along as guards, so a change that narrows the polytope while corrupting its edge directions or its joint frequencies is rejected rather than banked. The held-out quantities of App. J, per-respondent AUC, calibration, and objective-RR fidelity, are by contrast reported, not optimised against: because the loop minimises only literature coherence, the discrimination the network carries is an out-of-objective evaluation rather than a fitted target, which is why we read it as evidence about the literature-constrained CPTs (§6) and not as a number the method was tuned to. A crowdsourced deployment appends to the same log, each accepted contribution an update operator with a recorded before-and-after window, so the provenance of every tightening stays auditable.